

QuickEmbed: A 110M-Parameter Sentence Embedding Model for CPU Search

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Abstract

We present QuickEmbed, a 110-million-parameter sentence embedding model distilled from a 1-billion-parameter teacher. QuickEmbed encodes a search query in about 4 milliseconds on a single CPU core and keeps most of the teacher's retrieval quality on our benchmark.

1 Introduction

Semantic search needs an embedding for every query, and large embedding models need a GPU to answer quickly. Many search services run on CPUs only. We ask how small an embedding model can be before retrieval quality drops sharply.

2 Method

QuickEmbed is a 6-layer encoder trained by distillation: for 40 million sentence pairs it learns to reproduce the teacher's cosine similarities. Vectors have 384 numbers and are normalised to unit length, so a dot product is the cosine similarity.

3 Results

Our main result: QuickEmbed reaches a recall@10 of 88% on our 5,000-query benchmark, against 91% for the 1B-parameter teacher, while encoding a query in 4 ms on one CPU core instead of 60 ms.

Recall@10 is the share of queries whose relevant passage is among the ten closest vectors. The gap to the teacher is largest for queries containing numbers and product codes.

4 Limitations

English only, one benchmark built by the authors, and short passages of under 200 words. Queries with exact identifiers still need keyword search next to the embeddings.